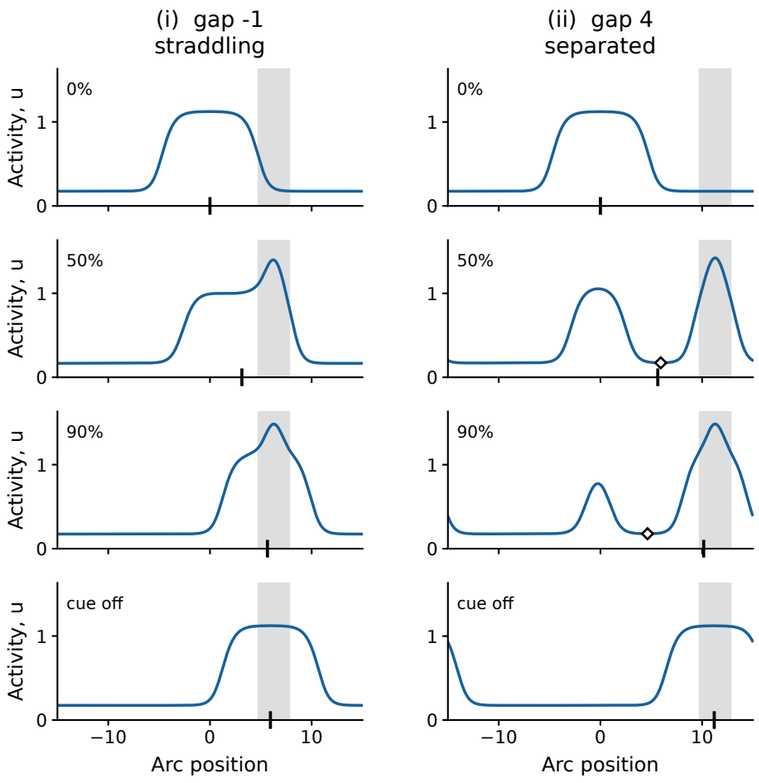
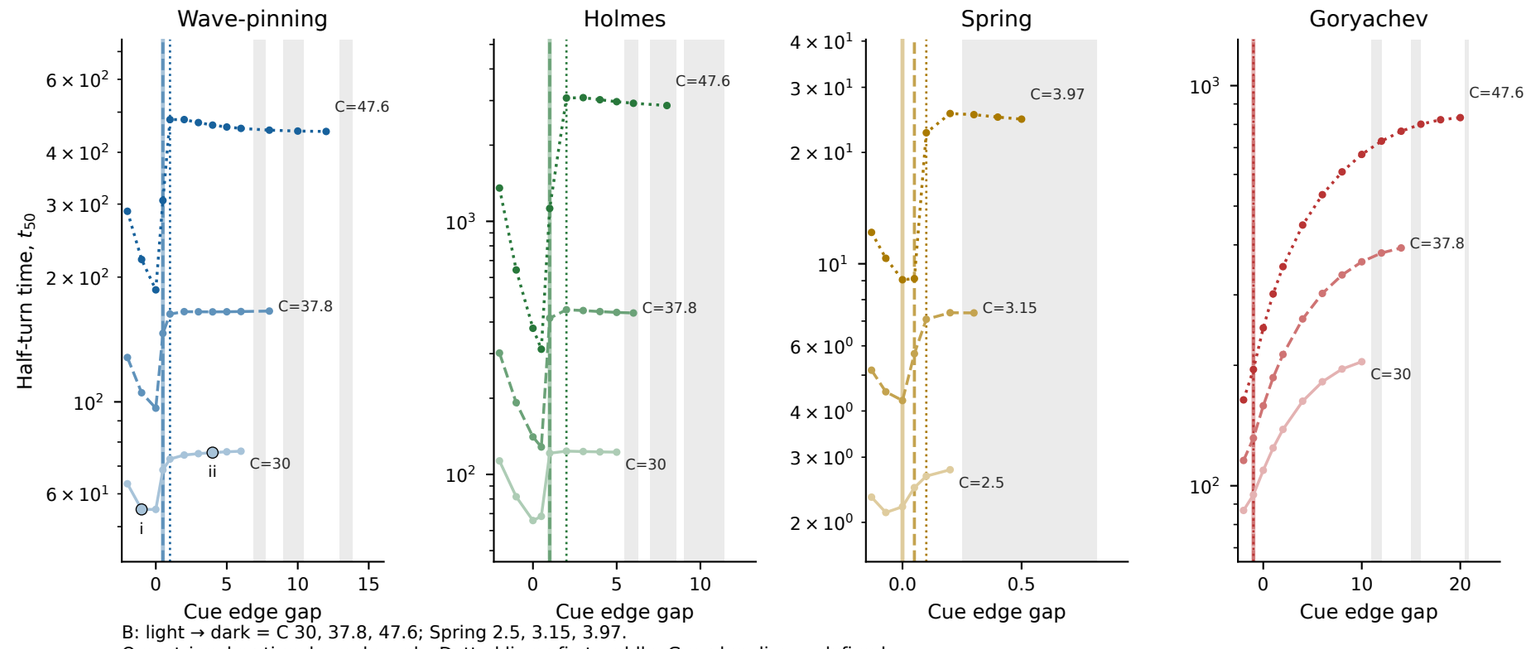
A

Grey: cue support; diamonds: saddle; | heading



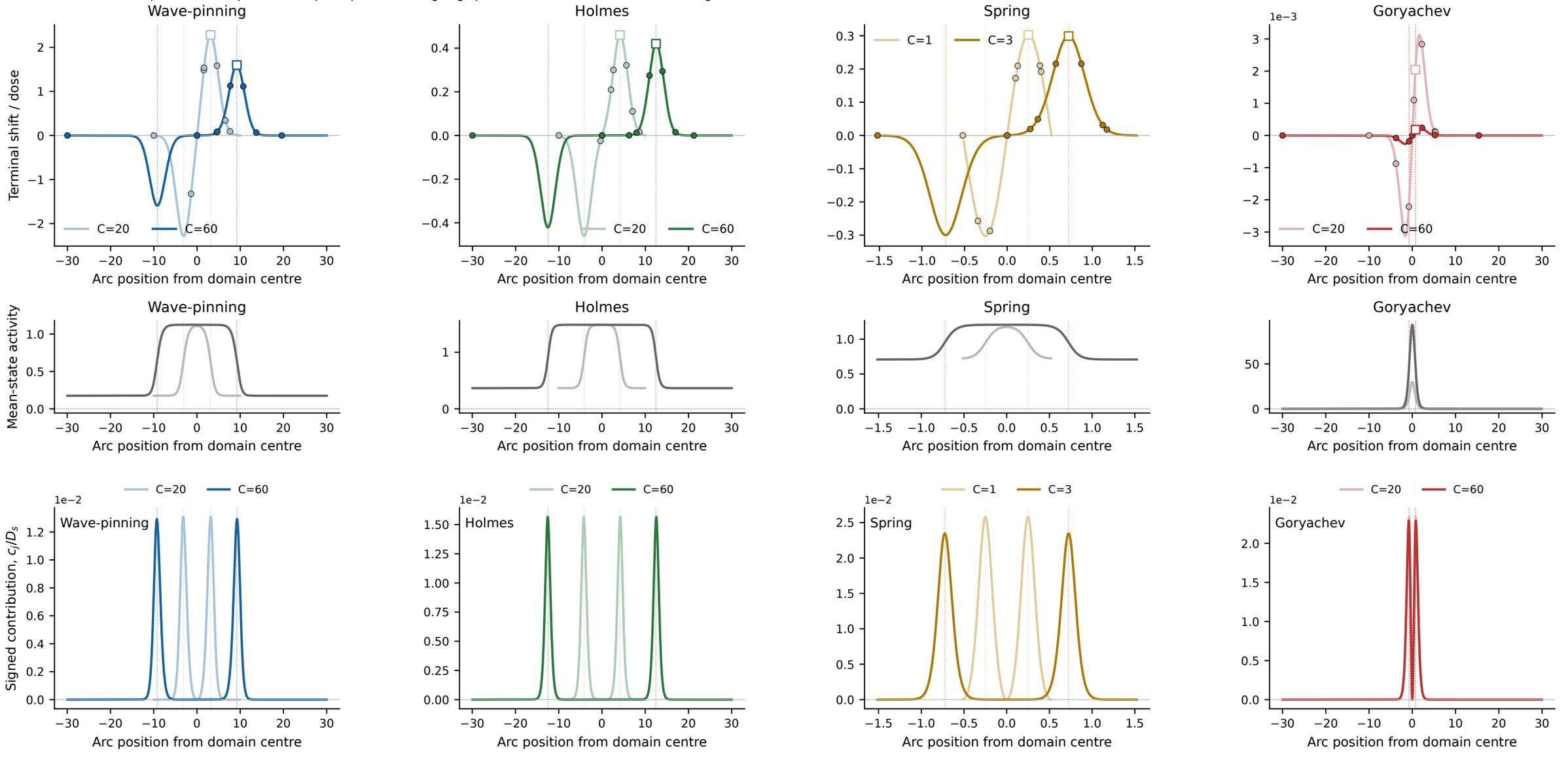
B



B: light → dark = C 30, 37.8, 47.6; Spring 2.5, 3.15, 3.97.
Open triangles: time lower bounds. Dotted lines: first saddle. Grey: heading undefined.

C

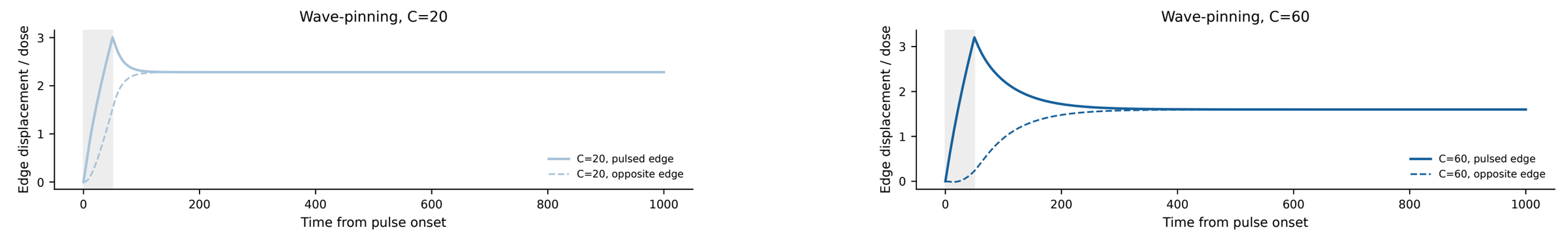
Lines: native map; circles: new pulse tests; open squares: existing edge pulse. Dotted verticals: half-contrast edges.



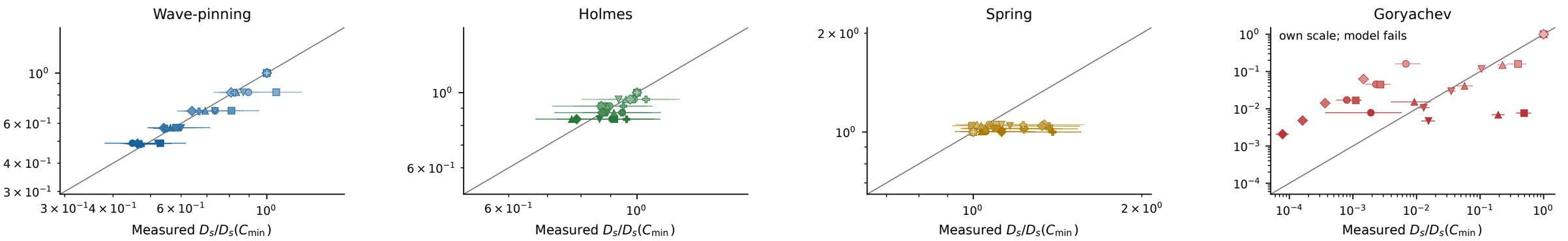
D

E

F



G



H

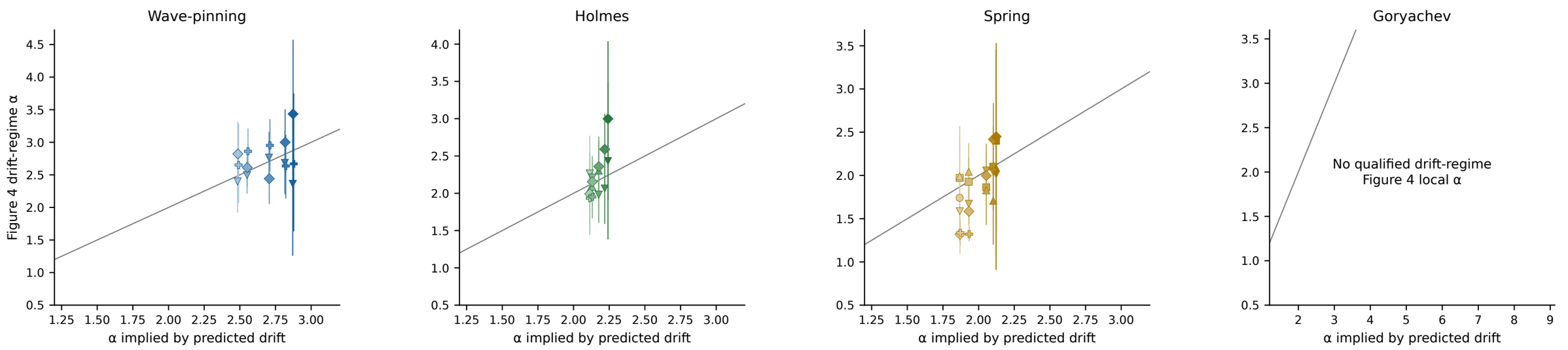


Figure 3. Enlargement slows both turning routes and selectively suppresses edge susceptibility

A. Wave-pinning activity profiles for a cue straddling the incumbent edge (i) or separated from it (ii), C=30. Rows show the initial state, half-turn, 90% turn and cue-free relaxation. Grey marks cue support, diamonds mark saddles and the black tick marks activity heading.

B. Enlargement slows half-turning through both incumbent sliding and replacement by a new region. Half-turn time is plotted against cue-edge gap for C=30, 37.8, 47.6 (Spring 2.5, 3.15, 3.97), light to dark. Vertical lines mark the first saddle at half-turn; grey marks undefined heading at one or more sizes. Open upward triangles with arrows denote time lower bounds.

C. Signed terminal displacement per dose for a Gaussian pulse centred at each arc position, at f=0.001: SD 1.5, duration 50 (Spring 0.15, 5). Light and dark are C20/C60 (Spring 1/3). Lines are the native linearized response to this pulse shape; circles are direct pulse tests and open squares mark the edge test. The edge pulse confirms the map locally; additional tests cover the domain centre, halfway to the edge, edge ±1 and ±3 pulse SD, quarter-basal arc and antipode. Maximum absolute gain error is 1.69% of map amplitude. The spatial response supplies the drift prediction in G.

D. Mean-state activity on the same centred physical arc coordinate as C and E. Dotted lines locate half-contrast edges. Spring uses material positions multiplied by mean circumference.

E. Nearly all predicted drift (99.6–100%) is generated within two input correlation lengths of the two edges, suggesting lateral-edge fluctuations as the cellular readout. Curves show signed per-position contributions divided by total predicted diffusion; negative values remain. Rounded shares per edge/plateau interior/basal arc (%) are: Wave-pinning C20: 50.0/<0.1/<0.1; Wave-pinning C60: 49.9/0.1/<0.1; Holmes C20: 50.0/<0.1/<0.1; Holmes C60: 50.0/<0.1/<0.1; Spring C1: 50.0/0/0; Spring C3: 49.8/0.2/0.2; Goryachev C20: 50.0/0/<0.1; Goryachev C60: 50.0/0/<0.1. Interior and basal zones exclude edge zones; exact signed fractions are supplied with the source data.

F. Wave-pinning C20 and C60: the pulsed edge (solid) and opposite edge (dashed) move together and retain a permanent shift after withdrawal. Grey marks pulse-on; displacement is divided by pulse dose. Largest/smallest terminal edge-gain folds are Wave-pinning 0.701, Holmes 0.914, Spring 0.989, Goryachev 0.088. Enlargement suppresses susceptibility strongly in Wave-pinning and Goryachev, while Holmes and Spring change little. Other time courses and integrated responses are supplementary.

G. A weak input anywhere shifts the domain by g(x) times dose; noise is many such inputs; drift follows from g and the input covariance; nothing is fitted. Shared-translation predicted versus measured diffusion is divided by each quantity's own smallest-size value at the same input (domain drift is also shown in Figure 4G). Grey is equality. Colours encode size; shapes encode f=0.0001, 0.0003, 0.001, 0.003, 0.005, 0.01 (circle, square, upward triangle, downward triangle, diamond, filled plus). Horizontal intervals are measured 95% bootstrap intervals. Wave-pinning's terminal gain fold gives $0.70^2 \approx 0.49$; the full spatial covariance predicts 0.490, versus 0.473 measured at f=0.001. The square relation is approximate because spatial shape and covariance also change. Goryachev fails quantitatively and retains its own scale. Absolute diffusion and the

Figure 3. Enlargement slows both turning routes and selectively suppresses edge susceptibility · continued

independent-edge candidate are supplementary.

H. The drift prediction implies a local size exponent, compared with measured heading-persistence α from Figure 4E. Predictions use the exact mean input of every observation; no f=0.001 extrapolation is used. The drift subset requires f≤0.01 and multiple-domain fraction≤0.05 throughout the fixed size neighbourhood. Native stationary-state and pulse-linearity gates must also pass. Grey is equality; vertical intervals are observed 95% bootstrap intervals; colours and shapes match G. 53 qualified pairs include all 15 Wave-pinning observations at f=0.003, 0.005 and 0.01. Its predicted α spans 2.48–2.88, above the size-independent-drift expectation of 2. Goryachev has no qualified pair. Every excluded condition and reason remains in the source data.

At the right edge, last tested linear peak amplitudes from the smallest to largest size are Wave-pinning 0.01024→0.00512 (0.5-fold), Holmes 0.0256→0.0256 (1-fold), Spring 0.0512→0.0512 (1-fold), and Goryachev 0.16384→0.65536 (4-fold). The next doubling exceeds 10% gain departure. Full brackets, all-size response curves and fixed-dose duration tests are in the methods and supplement.

$$G(x_0) = \sum_j g_j h(x_j - x_0) \tag{3.1}$$

The main map is the standard Gaussian-pulse response per dose, not raw cell-input g.

$$c_j = g_j \sum_k g_k \int_0^\infty \overline{K_u(j,k;t,t+v)}^{\,t}\, \mathrm{d}v, \qquad D_s = \sum_j c_j \tag{3.2}$$

Signed per-position contributions retain the exact native input covariance.

$$\alpha_{\mathrm{pred}} = 2 - \frac{\mathrm{d}\ln D_s}{\mathrm{d}\ln \bar{C}} \tag{3.3}$$

Same fixed adjacent-size neighborhoods as Figure 4E; exact matched-input predictions and explicit drift-regime eligibility.